📑 Basic Information:

Experiment No: 09

Date:

Title: Define a package number and in that define Roman class and implement Roman To Integer() and import the method in another class.

🎯 Objective /Aim:

To convert a Roman numeral into its integer value by calling the roman To Integer() method from the number.Roman class, then display the result.

Input: "LVIII" Output: 58

Explanation: L = 50, V= 5, III = 3.

🛠️ System Requirements:

Hardware:

-> Standard PC/Laptop

-> Intel Core i3/i5 processor or equivalent

-> Minimum 4 GB RAM

-> Minimum 500 MB free

Software:

->Operating System: Windows / Linux / macOS

-> Java Development Kit (JDK 17 or JDK 21)

-> Text Editor or IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code

-> Command Prompt / Terminal

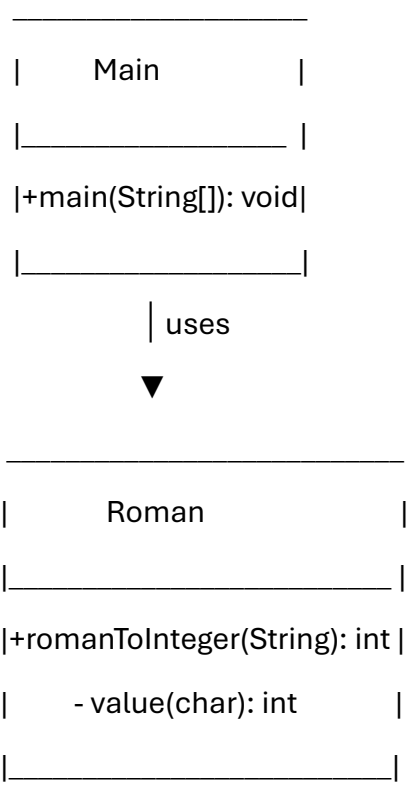
Algorithm/Logic Description:(For main program)

1. Start.
2. Store the Roman numeral "LVIII" in a string.
3. Call Roman.romanToInteger() and pass the string.
4. Store the returned integer.
5. Display the Roman numeral and its integer value.
6. Stop.

Algorithm/Logic Description:(For Roman class)

1. Start with the input Roman numeral and set result to 0.
2. Read each character from left to right.
3. Find the integer value of the current character.
4. If the current value is less than the next character's value, subtract it from result.
5. Otherwise, add it to result.
6. Repeat until all characters have been processed.
7. Return result.
8. Stop.

UML Class Diagram:



Source Code:(for main)

```
import number.Roman;
// Main class that demonstrates Roman numeral conversion
public class Main {
   public static void main(String[] args) {
      // Roman numeral to convert
```

```
    String input = "LVIII";
    // Convert the Roman numeral to an integer
    int result = Roman.romanToInteger(input);
    // Display the Roman numeral and its integer value
    System.out.println("Roman Number: " + input);
    System.out.println("Integer Value: " + result);
  }
}
```

Source Code:(for main)

```
package number;
public class Roman {
  // Converts a Roman numeral string to an integer
  public static int romanToInteger(String s) {
    int result = 0;
    // Process each Roman numeral character
    for (int i = 0; i < s.length(); i++) {
      int current = value(s.charAt(i));
      // Subtract the current value when a smaller numeral comes before a larger one
      if (i + 1 < s.length() &&
        current < value(s.charAt(i + 1))) {
        result -= current;
      } else {
        // Otherwise, add the current value
        result += current;
      }
    }
    return result;
  }
  // Returns the integer value for a Roman numeral character
```

```
    private static int value(char ch) {
        switch (ch) {
            case 'I': return 1;
            case 'V': return 5;
            case 'X': return 10;
            case 'L': return 50;
            case 'C': return 100;
            case 'D': return 500;
            case 'M': return 1000;
            // Return 0 for any character that is not a recognized Roman numeral
            default: return 0;
        }
    }
}
```

Relationship:

- Main uses the Roman class by calling its static roman To Integer() method.
- Roman To Integer() uses the private value() method to convert each Roman numeral character to its integer value.
- Main and Roman are separate classes; Main imports Roman from the number package.

Output Screenshot:



Result:

The Main class uses the Roman class in the number package to convert a Roman numeral into its integer equivalent. For the given input, LVIII is converted to 58.